Water Quality Prediction: Energy-Aware Offline Multimodal Neuromorphic Edge Inference ⋆

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Abstract: This work presents an Edge-AI-centered neuromorphic pipeline for multimodal water-quality prediction in a remote Amazonian setting. The central objective is fully offline deployment on constrained hardware, avoiding cloud dependence under limited connectivity and power budgets. A spiking neural network (SNN) is used to jointly predict pH, dissolved oxygen, and turbidity from georeferenced physicochemical inputs collected in a field campaign of 60 georeferenced samples. To improve stability under small-data conditions, we apply controlled Gaussian augmentation (expanding the training support tenfold, to 600 samples) and optimize the model through systematic grid search over temporal steps, learning rate, threshold, decay, and hidden-layer size. The resulting multimodal SNN is evaluated with multi-seed protocols reported with 95% confidence intervals and nonparametric significance tests (exact Wilcoxon signed-rank and Friedman) and then deployed on heterogeneous edge platforms (ESP32, Raspberry Pi, and Android devices). We report throughput, latency, and incremental energy-per-inference using idle/load measurements, including INA219-based telemetry where applicable. Results show that although Random Forest and LSTM can achieve lower average macro error, the SNN remains competitive and operationally robust for edge scenarios, while enabling practical offline inference across microcontroller-, SBC-, and smartphone-class hardware. Overall, the study demonstrates that neuromorphic Edge AI is a viable strategy for autonomous environmental monitoring in connectivity- and energy-constrained regions.

Keywords: Spiking Neural Networks; Edge AI; Energy Benchmarking; Water Quality Prediction; Multimodal Modeling; Offline Inference.

1. INTRODUCTION

Water-quality monitoring in remote Amazonian areas is constrained by sparse infrastructure, high field logistics cost, and intermittent network connectivity. Edge AI deployment is the central motivation of this work: local, fully offline inference is required to keep monitoring functional under unstable communications and limited energy budgets. In this context, continuous dependence on cloud inference is operationally fragile and energetically expensive, especially for battery-powered deployments and long-duration campaigns. At the same time, water-quality variables such as pH, dissolved oxygen (OD), and turbidity are highly relevant for environmental screening and public-health risk mitigation.

Conventional machine-learning pipelines can provide strong predictive performance, but many are designed for server-centric execution and do not directly address energy constraints at the edge. Neuromorphic models based on spiking neural networks (SNNs) offer an event-driven alternative compatible with low-power embedded systems, yet practical evidence in real environmental sensing workflows remains limited, particularly when combining multimodal prediction and platform-level energy benchmarking.

⋆ Financial support from Eldorado Research Institute, Motorola Mobility Comércio de Produtos Eletrônicos Ltda, Flextronics da Amazônia Ltda, and the PDI project "Mob4Ai – Advanced Artificial Intelligence for Neuromorphic Mobile Systems" (Agreement No. 008/2024, under Decree No. 10.521/2020 and Federal Law No. 8.387/1991) is gratefully acknowledged.

This paper proposes and evaluates an offline multimodal SNN pipeline for water-quality prediction in an Amazonian field scenario, including end-to-end deployment and cross-platform energy analysis. The study integrates data processing, model training, geospatial interpretation, and embedded inference on heterogeneous hardware (ESP32, Raspberry Pi, and Android devices).

The main contributions of this work are fourfold: (i) the development of a multimodal SNN-based water-quality prediction pipeline designed for offline edge deployment; (ii) the creation of an augmented georeferenced dataset derived from field measurements in Lago do Serpa; (iii) a comparative latency–energy evaluation across heterogeneous platforms, including ESP32, Raspberry Pi, and Android devices; and (iv) a statistically grounded evaluation protocol for small-data environmental regression, combining repeated-seed benchmarking, confidence intervals, and nonparametric significance testing. Beyond energy benchmarking, the scientific contribution lies in demonstrating, end to end, that multimodal neuromorphic regression of physicochemical water-quality targets is feasible from field acquisition to fully offline inference on three distinct classes of edge hardware — a combination not yet reported for freshwater monitoring in the Amazon region.

2. RELATED WORK

2.1 Water Quality Prediction

Data-driven prediction of water-quality indicators has been widely explored with ANN, tree-based models, and recurrent architectures. Reviews and recent studies report consistent use of physicochemical variables and geospatial context for forecasting pH, OD, turbidity, and composite indicators Yan et al. (2024); Ahmed et al. (2019); Al-Adhaileh and Alsaade (2021); Pan et al. (2025). LSTM-based approaches are common where temporal records are dense Liu et al. (2019); Pyo et al. (2023); Qi et al. (2020), while hybrid and ensemble methods improve robustness under nonlinear behavior and limited sample regimes Wu and Wang (2022); Chadalavada et al. (2025); Kalaivanan and Vellingiri (2022); Palani et al. (2008).

Beyond model choice, an important distinction in the literature is the operational objective: maximum predictive accuracy in centralized infrastructure versus robustness under noisy sensing, sparse sampling, and field constraints — a critical distinction for inland waters, where model behavior is affected by local heterogeneity, seasonality, and measurement uncertainty Yan et al. (2024); Lowe et al. (2022); Rahu et al. (2023).

2.2 Edge AI and Energy Evaluation

For environmental IoT deployments, inference cost must be evaluated jointly with predictive quality. Prior work highlights the relevance of platform-level benchmarking and hardware-aware model design Rosero-Montalvo et al. (2020); Zulkifli et al. (2022); Lamrini et al. (2024); El-Shafeiy et al. (2023). Recent neuromorphic and energy-efficient AI studies indicate that event-driven models can be attractive for constrained settings, but deployment evidence across heterogeneous edge hardware is still developing Aral (2024); Azadi et al. (2025); Huijbregts et al. (2024); Aribe Jr. (2025). This gap motivates the present work: a unified comparison of offline multimodal inference quality and energy-per-inference behavior on multiple practical platforms.

Most prior studies report either algorithmic metrics or hardware metrics in isolation. In contrast, field deployment requires a coupled view of latency, throughput, and incremental energy under realistic idle/load transitions. Consequently, cross-platform benchmarking should be interpreted as a systems result, not only as a model result.

Positioning this work against recent neuromorphic environmental-monitoring studies clarifies its novelty. Aral (2024) discuss the promise of neuromorphic Edge AI for rural monitoring at a conceptual level, without field deployment; Azadi et al. (2025) demonstrate embedded neuromorphic inference for industrial IoT, but not for water-quality regression; Huijbregts et al. (2024) address SNN efficiency at the circuit level; and Al-Shareeda et al. (2025) deploy ESP32-based water monitoring without neuromorphic models. To our knowledge, no prior study combines multimodal SNN regression of physicochemical targets, field data from an Amazonian lake, and measured energy-per-inference across microcontroller-, SBC-, and smartphone-class hardware in a single offline pipeline, which is the gap addressed here.

3. DATASET DESCRIPTION

The field campaign was conducted in a freshwater environment in the Amazon region (Itacoatiara, AM, Brazil), with georeferenced sampling points distributed to capture local spatial gradients. The dataset includes position in UTM 21S and physicochemical variables used as model inputs and outputs, and is publicly available in Mendeley Data Vasconcelos Assunção et al. (2026). The spatial coverage of the campaign is shown in Fig. 1, and summary statistics of the prediction targets are reported in Table 2.

Data collection followed a spatially distributed design to represent different lake micro-zones instead of repeatedly sampling a single coordinate, supporting spatial interpretation and local decision support (interpolation maps and composite screening indices). Measurements were consolidated into a tabular structure suitable for edge-oriented inference and reproducible training.

3.1 Variables and Acquisition

Input features were x, y, temperature (TEMP), apparent color in amber scale (COR_A), and visual color index (COR_V). Target variables were pH, dissolved oxygen (OD), and turbidity (TURB). The original field dataset comprises 60 georeferenced samples. The sampling design prioritized spatial representativeness over dense temporal repetition, consistent with the objective of generating interpretable spatial surfaces.

Table 1 reports the Pearson correlation between each input feature and each target in the original 60-sample dataset. Spatial coordinates and color indices are strongly associated with pH ($r = -0.85$ for y; $r = -0.79$ for COR_A) and turbidity ($r = 0.69$ and $r = 0.68$, respectively), indicating exploitable spatial and optical gradients. In contrast, OD

Table 1. Pearson correlation between input features and targets (original dataset, $n = 60$).

Feature	pH	OD	TURB
x	0.75	−0.23	−0.67
y	−0.85	0.18	0.69
TEMP	−0.56	0.24	0.33
COR_A	−0.79	0.05	0.68
COR_V	−0.46	0.08	0.50

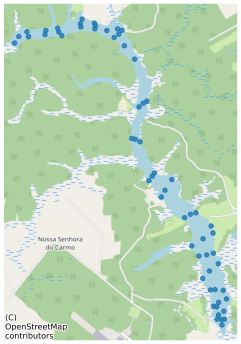


Figure 1. Sampling locations over OpenStreetMap basemap (UTM 21S).

Table 2. Target summary statistics in the original field dataset.

Variable	Mean	Std	Min	Max
pH	5.73	0.31	5.15	6.50
OD (mg/L)	1.86	0.32	1.26	2.52
Turbidity (NTU)	2.77	1.37	0.06	6.18

is weakly correlated with all available inputs ($|r| \leq 0.24$), which anticipates the higher difficulty of predicting this target with the current feature set.

3.2 Data Augmentation

Because the original dataset is small ($n = 60$), synthetic augmentation was applied with factor ×10 using light Gaussian perturbation ($\sigma = 0.01$) in normalized space, yielding 600 training-support samples. This strategy was used to stabilize training and reduce split sensitivity, while preserving marginal distributions and physical plausibility. The augmented set is used for training support only and does not represent additional field measurements. Gaussian perturbation was adopted for its transparency and low distortion risk in this regime; alternative augmentation strategies for small tabular datasets — such as SMOTE-style interpolation for regression, mixup, and generative approaches (e.g., copula- or GAN-based synthesis) — are natural candidates for a systematic comparison in future work.

To minimize distortion risk, augmentation was constrained to low-amplitude perturbations and evaluated with distributional checks and ablation analysis: the goal was to increase local density around observed samples and improve optimization stability across randomized splits, not to create synthetic dynamics absent from the field campaign.

Finally, the dataset should be interpreted as an initial regional benchmark rather than a universal hydrological representation. The current sampling supports comparative modeling and deployment analysis, while broader external generalization requires multi-season campaigns and additional freshwater sites with different hydrodynamic and anthropogenic profiles. The small sample size imposes concrete methodological limits: absolute metric values carry wide uncertainty, small differences between models may not be stable across resamplings, and single-season sampling cannot capture hydrological seasonality. These limits motivated the conservative protocol adopted here (repeated seeds, confidence intervals, and nonparametric tests) and bound the scope of the conclusions to comparative and deployment-oriented claims.

4. METHODOLOGY

4.1 Multimodal Input Representation

To avoid ambiguity between symbolic variables and sensor names, we represent each observation using an ordered multimodal feature vector:

$$\mathbf{x} = [x, y, \mathrm{TEMP}, \mathrm{COR}_A, \mathrm{COR}_V] \in \mathbb{R}^5, \tag{1}$$

and a three-output target vector:

$$\mathbf{y} = [\mathrm{pH}, \mathrm{OD}, \mathrm{TURB}] \in \mathbb{R}^3. \tag{2}$$

Input features are normalized before spike encoding. We then apply rate coding so that each normalized scalar is converted into a spike train over T time steps, with firing probability proportional to its magnitude. This produces a temporal representation compatible with the downstream LIF dynamics while preserving the multimodal structure of the original sample.

4.2 SNN Architecture

The model follows a feed-forward SNN with topology $5 \to 64 \to 3$, with a hidden leaky integrate-and-fire (LIF) layer, as illustrated in Fig. 2. The membrane update is

$$\mathbf{u}_{t+1} = \beta\,\mathbf{u}_t + \mathbf{W}\mathbf{s}_t - \theta\,\mathbf{z}_t, \tag{3}$$

where β is membrane decay, θ is firing threshold, and $\mathbf{z}_t$ are emitted spikes. Output prediction is computed by temporal aggregation across steps.

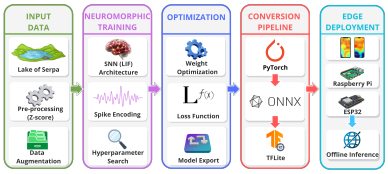


Figure 2. Proposed multimodal SNN architecture for offline edge inference.

Figure 2 presents the complete end-to-end workflow: field data preparation (normalization and controlled augmentation), neuromorphic training with LIF neurons and hyperparameter search, model optimization and export through the PyTorch→ONNX→TFLite interoperability path, and offline edge deployment on Android, Raspberry Pi, and ESP32.

4.3 Training Protocol

Training used Adam optimizer and weighted MSE loss with early stopping. Hyperparameters were selected by grid search over hidden size, learning rate, temporal steps T, threshold, and decay factor. The full search space was: hidden size $\in \{32, 64, 96\}$, $T \in \{10, 15, 25\}$, learning rate $\in \{0.0005, 0.001, 0.002\}$, decay $\beta \in \{0.90, 0.95\}$, threshold $\theta \in \{0.8, 1.0\}$, and augmentation noise $\sigma \in \{0.01, 0.02, 0.03\}$, totaling 324 configurations evaluated by validation loss. The selected configuration was hidden size 64, $T = 15$, learning rate 0.002, $\beta = 0.95$, $\theta = 0.8$, and $\sigma = 0.01$. Evaluation used repeated random seeds and macro metrics (MAE, RMSE, and R^2) over the three targets. The ten independent seeded runs implement repeated random subsampling validation, which approximates cross-validation behavior in this small-data regime; a full k-fold protocol is planned for the extended dataset in future campaigns.

4.4 Hyperparameter Sensitivity

The grid results also provide a sensitivity analysis of the LIF parameters and the temporal encoding. Averaging validation loss marginally over the 324 configurations, learning rate had the strongest effect (mean loss 0.826 at 0.0005 vs. 0.694 at 0.002). The LIF threshold showed a consistent preference for $\theta = 0.8$ over $\theta = 1.0$ (0.747 vs. 0.769), while the decay factor had a minor influence (0.754 at $\beta = 0.90$ vs. 0.762 at $\beta = 0.95$). The temporal parameter T had a weak effect on accuracy (mean loss 0.766, 0.760, and 0.748 for $T = 10, 15, 25$), but inference cost grows approximately linearly with T, since each additional step implies one more forward pass of spike propagation. Hence T is primarily an energy–latency knob rather than an accuracy knob in this task, and $T = 15$ was retained as a balanced operating point.

As shown in Fig. 3, training and validation losses decrease rapidly in the initial epochs and converge to a stable plateau with limited divergence, indicating consistent optimization without strong overfitting and supporting the use of early stopping in this small-data regime.

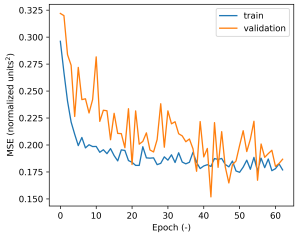


Figure 3. Training and validation loss (MSE) of the selected multimodal SNN.

5. ENERGY MEASUREMENT SETUP

5.1 Instrumentation and Procedure

ESP32 and Raspberry Pi: current and voltage were sampled every 500 ms using INA219. Two windows were captured: idle (N_{idle} samples) and load (N_{load} samples) during benchmark execution.

Table 3. Android device specifications (manufacturer data; RAM in parentheses is the OS-reported MemTotal).

	Android A	Android B	Android C
Device	Redmi Note 12	Galaxy A56 5G	Redmi Note 11S
Model	23021RAAEG	SM-A566E	2201117SG
SoC	Snapdragon 685	Exynos 1580	Helio G96
Process	6 nm	4 nm	12 nm
CPU	4×A73 2.8	1×A720 2.9	2×A76 2.05
(GHz)	4×A53 1.9	3×A720 2.6	6×A55 2.0
		4×A520 1.95	
GPU	Adreno 610	Xclipse 540	Mali-G57 MC2
RAM	6 GB (5.77)	8 GB (7.60)	6 GB (5.86)
Android	15	16	13

Table 4. Measured setup and sampling windows by platform.

Platform	Source	N_{idle}	N_{load}
ESP32 (MicroPy)	INA219	50	50
Raspberry Pi 4B	INA219	97	440
Android (A-C)	System telem.	200	200

Android devices: current and voltage were read from system APIs under controlled idle/load phases. The complete instrumentation and sampling windows are summarized in Table 4.

Incremental power was estimated as $\Delta P = P_{load} - P_{idle}$, and energy per inference as $E_{inf} = \Delta P / \text{throughput}$.

Table 3 details the three Android devices (A–C) used in the campaign. Since inference runs on the CPU through the TFLite interpreter (XNNPACK, no GPU/NNAPI delegate), the CPU cluster configuration and available memory are the resources relevant to the AI workload; GPUs are listed for completeness only.

The resulting deployment envelope combines throughput and incremental energy per inference: high-rate scenarios favor smartphone and SBC execution, whereas low-rate autonomous sensing can still be supported on microcontroller-class hardware.

6. EXPERIMENTAL RESULTS

6.1 Prediction Performance

Table 5 summarizes macro performance in a multi-seed benchmark ($n = 10$), reporting mean and standard deviation. Random Forest and LSTM obtained the lowest average errors, while the proposed SNN remained competitive and substantially more stable than the MLP baseline.

Table 5. Multi-seed multimodal prediction performance (macro metrics).

Model	MAE macro	RMSE macro	R^2 macro
Random Forest	$\mathbf{0.409 \pm 0.062}$	0.557 ± 0.086	$\mathbf{0.335 \pm 0.131}$
LSTM	0.428 ± 0.059	$\mathbf{0.548 \pm 0.073}$	0.313 ± 0.139
SNN (proposed)	0.447 ± 0.064	0.566 ± 0.072	0.291 ± 0.146
MLP	0.781 ± 0.193	0.983 ± 0.178	-5.460 ± 5.220

Although Random Forest achieved the lowest MAE, the SNN demonstrated competitive predictive performance while enabling deployment on low-power edge devices,

which is essential for remote environmental monitoring scenarios. The ranking indicates that the SNN is not the lowest-error model in absolute terms, but it remains in a competitive range with materially better robustness than MLP in this small-data regime.

To quantify uncertainty, 95% confidence intervals (t-distribution, $n = 10$ seeds) were computed for all macro metrics. For MAE macro: SNN $[0.399, 0.495]$, Random Forest $[0.362, 0.456]$, LSTM $[0.384, 0.472]$, and MLP $[0.636, 0.926]$; for R^2 macro: SNN $[0.181, 0.401]$, Random Forest $[0.237, 0.434]$, LSTM $[0.209, 0.418]$, and MLP $[-9.396, -1.524]$. The intervals of SNN, Random Forest, and LSTM overlap substantially, whereas MLP is clearly separated.

Complementing the sign test reported in the original submission, exact Wilcoxon signed-rank tests over the ten paired seeds show no significant difference between SNN and Random Forest (MAE $p = 0.064$; RMSE $p = 0.846$; R^2 $p = 0.922$) or between SNN and LSTM (MAE $p = 0.105$; RMSE $p = 0.131$; R^2 $p = 0.232$), while SNN significantly outperforms MLP on all metrics ($p = 0.002$). A Friedman test across the four models is significant (MAE: $\chi^2(3) = 19.56$, $p < 0.001$), with mean ranks of 1.7 (Random Forest), 1.9 (LSTM), 2.4 (SNN), and 4.0 (MLP), confirming that the global difference is driven by MLP rather than by separation among the three leading models.

6.2 Per-Target Performance

Table 6 details per-target behavior. All three leading models achieve good pH prediction ($R^2 \approx 0.67$–0.72) and comparable turbidity error, while OD exhibits negative mean R^2 for every model, including Random Forest. This is consistent with the correlation analysis of Table 1: OD is weakly correlated with all available input features, so the difficulty is attributable to the feature set rather than to the choice of model.

Table 6. Per-target MAE and R^2 (mean over 10 seeds); cells show MAE / R^2.

Model	pH	OD	TURB
SNN (proposed)	0.125 / 0.670	0.204 / −0.162	1.012 / 0.364
Random Forest	0.097 / 0.715	0.203 / −0.063	0.926 / 0.354
LSTM	0.112 / 0.673	0.203 / −0.154	0.968 / 0.422
MLP	0.777 / −13.39	0.352 / −3.021	1.214 / 0.032

6.3 Augmentation Ablation

To quantify augmentation impact, we compared SNN training with no augmentation (factor 1) and augmentation ×10 (Gaussian noise, $\sigma = 0.01$) under the same protocol across five seeds (7, 13, 21, 42, 84). Augmentation reduced macro MAE from 0.496 ± 0.060 to 0.483 ± 0.075 (2.53% reduction), with slight macro R^2 improvement ($0.254 \rightarrow 0.256$) and nearly unchanged macro RMSE ($0.615 \rightarrow 0.617$). The most relevant per-target gain was for turbidity MAE ($1.139 \rightarrow 1.102$), supporting augmentation as a practical stabilization strategy for this dataset size.

6.4 Target-Weight Sensitivity

Because turbidity is the most difficult target, we also evaluated a small loss-weight sweep in the SNN to quantify the trade-off between global fit and TURB-focused optimization. Table 7 shows that increasing TURB weight can reduce MAE_{TURB}, but with a slight decrease in global R^2.

Table 7. SNN loss-weight sweep (10 seeds).

Weights (pH, OD, TURB)	MAE macro	MAE TURB	R^2 macro
(0.20, 0.20, 0.60)	**0.447**	1.012	**0.291**
(0.15, 0.15, 0.70)	0.452	1.022	0.289
(0.10, 0.10, 0.80)	0.454	1.028	0.288
(0.05, 0.15, 0.80)	0.448	**1.006**	0.280

6.5 Predicted-vs-True Behavior

Figure 4 complements aggregate metrics by showing point-wise agreement between predicted and observed values for each target. pH and OD concentrate closer to the identity line, while turbidity presents larger spread, consistent with its higher error contribution in previous tables. For OD in particular, the visual agreement should be read together with Table 6: the low variance of OD in the dataset (Table 2) keeps absolute errors small, but the weak input–target correlation limits explained variance for all evaluated models, not only the SNN.

6.6 Spatial Results and Pollution Index

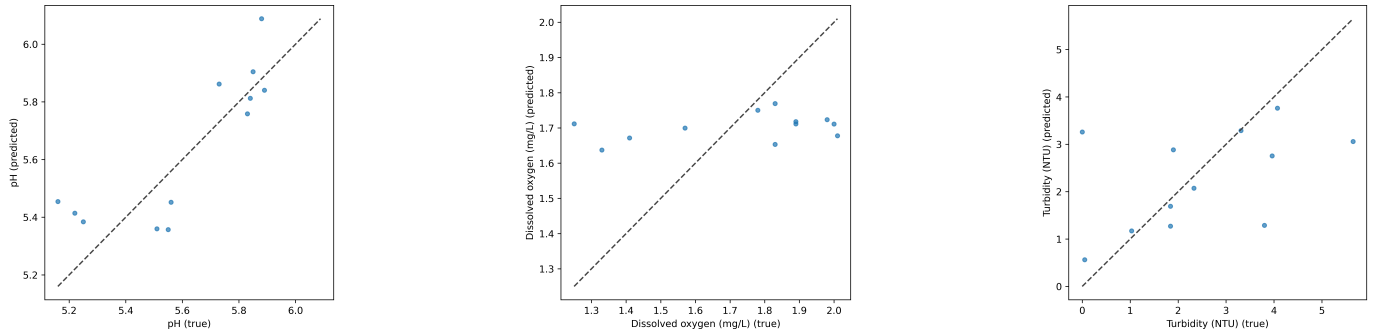
IDW interpolation was used to convert point predictions into continuous spatial surfaces, improving environmental interpretation beyond point-wise metrics. As shown in Figure 5, the resulting maps highlight nonuniform spatial behavior across pH, dissolved oxygen, and turbidity, and indicate candidate micro-zones for prioritized field inspection.

To support operational screening, a composite pollution index was computed from multimodal predictions and normalized to $[0, 1]$, where higher values indicate worse relative condition within the analyzed set. As illustrated in Figure 6, the point-level representation and its IDW spatial estimate over the OSM basemap provide a direct visual tool for identifying critical areas.

6.7 Energy Consumption and Latency

Table 8 reports incremental energy-per-inference and throughput across deployment platforms. ESP32 is feasible for low-rate autonomous scenarios but has higher energy per inference due to limited throughput. Raspberry Pi and Android devices provide much lower mJ/inf in high-throughput settings.

Figure 7 compares the incremental energy per inference across platforms on a logarithmic scale: ESP32 has the highest energy cost per prediction, mainly due to its low throughput, while Raspberry Pi and Android devices operate in a much lower mJ/inf range. Among mobile platforms, higher throughput does not always imply lower energy per inference, reinforcing that platform selection



(a) pH (b) OD (c) TURB

Figure 4. Predicted vs. true values in the multimodal SNN for pH, dissolved oxygen, and turbidity. (Figure enlarged to full page width in this revision for axis readability.)

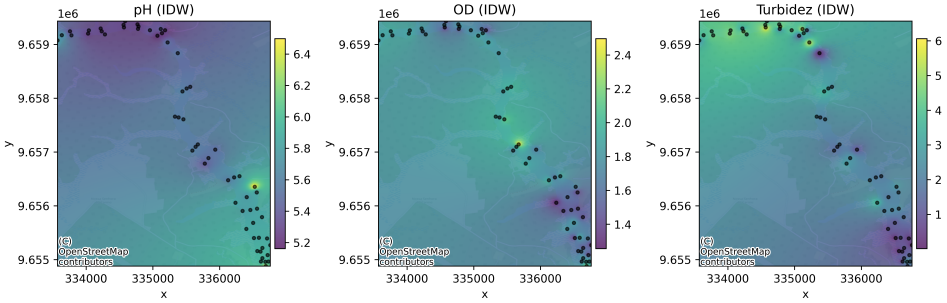


Figure 5. IDW interpolated surfaces for pH, dissolved oxygen, and turbidity from sampled points. (Figure enlarged to full page width in this revision for axis readability.)



(a) Pollution index at sample points. (b) IDW surface of the index.

Figure 6. Spatial estimate of the composite pollution index from multimodal predictions (normalized scale $[0, 1]$) over an OpenStreetMap basemap (UTM 21S). (Figure enlarged to full page width in this revision for axis readability.)

Table 8. Latency/throughput and incremental energy across platforms.

Platform	Thr. (samples/s)	ΔP (W)	E (mJ/inf)
ESP32 (MicroPy)	5.0	0.056	11.204
Raspberry Pi 4B	4670.50	0.239	0.0512
Android A	12337.40	0.128	0.0104
Android B	30905.08	0.468	0.0151
Android C	16121.71	0.046	0.00285

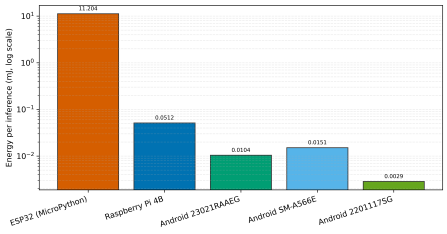


Figure 7. Comparative energy per inference (log scale). (Figure enlarged in this revision for axis readability.)

should jointly consider processing speed and incremental power behavior.

Figure 8 summarizes the multi-criteria deployment envelope by placing throughput, inverse latency, inverse energy,

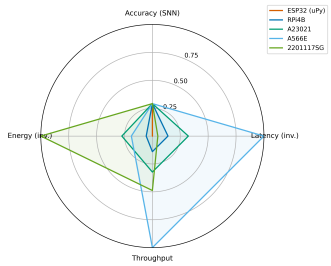


Figure 8. SNN deployment comparison across platforms using normalized criteria (throughput, inverse latency, inverse energy, and predictive reference). (Figure enlarged in this revision for axis readability.)

and predictive reference in the same frame, and reinforces that no single platform dominates all objectives simultaneously. Android devices and the Raspberry Pi occupy a broader envelope on response-related axes, favoring denser temporal sampling and interactive field workflows; ESP32 contracts on those axes but remains relevant for sparse, autonomous, fully offline operation. The figure should therefore be read as a deployment map rather than as a universal ranking.

7. DISCUSSION

Results show a clear trade-off between predictive accuracy and deployment constraints. In the multi-seed benchmark, Random Forest and LSTM reached lower average macro errors, while the SNN remained competitive and clearly more stable than MLP. Wilcoxon and Friedman outcomes, together with the overlapping confidence intervals, indicate that Random Forest, LSTM, and the SNN are statistically indistinguishable on this dataset, which reinforces that the SNN should be interpreted as a robust edge-oriented alternative rather than a universal best-accuracy model. From a methodological standpoint, the multimodal setup proved valuable because it keeps physically related targets in a single inference pipeline and simplifies downstream decision logic in field applications.

From the systems perspective, platform behavior is heterogeneous: Device B reached the highest throughput, Device C achieved the lowest incremental energy per inference, and ESP32 provided full offline feasibility at much lower processing rates. Since baseline platform power varies with architecture and operating-system behavior, energy per inference combined with throughput is a more actionable criterion than absolute watts alone. Augmentation ablation showed a modest but consistent macro-MAE improvement (2.53%) and turbidity gain, indicating that controlled synthetic expansion helps stabilize training in this low-sample regime.

7.1 Practical Deployment Guidance

The combined results allow a simple deployment guideline for field teams: for dense temporal sampling, frequent map updates, or near-real-time dashboards, smartphone or SBC-class execution is preferable; for sparse periodic checks with strict hardware simplicity, ESP32 remains viable and operationally attractive despite higher mJ/inf values.

A second practical point is model-selection policy: in constrained field scenarios, selecting only by minimum MAE can be suboptimal. A better criterion is a multi-objective envelope combining acceptable prediction error, stable seed behavior, platform throughput, and incremental energy per inference, extensible with constraints such as battery budget and telemetry cadence.

The choice of the SNN over Random Forest deserves explicit justification, since Random Forest achieved the lowest mean MAE. First, the accuracy difference is not statistically significant: Wilcoxon tests and confidence intervals show full overlap between the two models. Second, the SNN offers properties that Random Forest does not: event-driven computation whose cost scales with spike activity, a single compact dense model that exports directly to ONNX/TFLite for all three platform classes, and forward compatibility with neuromorphic accelerators, where sparse spiking execution can reduce energy well below the CPU-based measurements reported here. Third, the temporal encoding parameter T provides a tunable accuracy–energy knob that has no counterpart in tree ensembles. The SNN is therefore selected not as the lowest-error model, but as the statistically equivalent model with the best deployment envelope.

Regarding Edge AI versus Cloud AI, the trade-off goes beyond raw compute. Cloud inference offers larger models and simpler updates, but in remote Amazonian deployments it adds radio energy that typically dominates the per-sample budget, latency and availability tied to intermittent links, recurring connectivity costs, and exposure of raw environmental data to third-party infrastructure. Fully offline edge inference inverts this profile: energy is bounded by local computation (measured here at μJ-to-mJ scale per inference), monitoring continues during outages, and only aggregated results need eventual synchronization. A hybrid pattern — local screening with opportunistic cloud consolidation — is a natural evolution of the present pipeline.

7.2 Error Interpretation and Validity

The predicted-vs-true plots indicate a clear target-dependent error profile: pH and OD follow the identity trend with lower dispersion, while turbidity exhibits wider spread. This behavior is coherent with freshwater heterogeneity and reinforces that turbidity should be interpreted with broader confidence margins in operational use. From an application perspective, this asymmetry supports a layered decision strategy: treat pH and OD as stronger short-term screening signals, and use turbidity primarily to flag candidate zones for additional sampling rather than as a standalone trigger. For dissolved oxygen specifically, the per-target analysis shows that no evaluated model achieves positive mean R^2, which points to a feature-set limitation rather than a modeling one: the current inputs carry little OD-related information ($|r| \leq 0.24$). Improving OD prediction will therefore require enriching the input modality — e.g., conductivity, oxidation-reduction potential, chloro-

phyll, or diel temperature profiles — rather than further model tuning.

Threats to validity remain centered on sample size and regional representativeness. Although augmentation, multi-seed evaluation, and cross-platform benchmarking improved robustness, broader generalization still requires multi-season and multi-site campaigns with harmonized acquisition protocols and repeated external validation. Transferability to other regions should be framed at the level of the pipeline rather than the trained weights: the learned spatial gradients are specific to Lago do Serpa, so direct application of the fitted model to another lake is not expected to preserve accuracy. What transfers is the methodology — feature design, augmentation policy, statistical protocol, and deployment stack — which can retrain locally from a comparably small campaign in a new site. Cross-site fine-tuning and domain-adaptation strategies are planned to quantify this transfer cost explicitly. Reproducibility is a practical strength of this pipeline: the modeling protocol, seed-based evaluation, and offline deployment measurements are explicit and portable, allowing direct replication in other low-connectivity contexts. Beyond predictive metrics, the Raspberry Pi deployment achieved sub-millisecond inference and the ESP32 validation confirmed autonomous offline execution under constrained resources, reinforcing that local inference is practical across heterogeneous edge profiles.

8. CONCLUSION

This paper presented an energy-aware offline multimodal SNN pipeline for water-quality prediction in an Amazonian edge-monitoring context. The proposed approach demonstrated practical feasibility from training to deployment, including geospatial interpretation and cross-platform latency/energy benchmarking. The results indicate that neuromorphic inference is deployable under real hardware constraints, with measurable platform-specific trade-offs between throughput and energy per inference. While not always the best in absolute MAE, the SNN remains a robust option when temporal event-driven processing and edge compatibility are first-class requirements.

Threats to validity should be explicitly considered when interpreting these findings. The main internal-validity risk is the limited size of the original field dataset, which can increase split sensitivity and make small metric gaps unstable across partitions; this was mitigated through repeated-seed evaluation, conservative augmentation, and per-target reporting, but not fully eliminated. External validity is also bounded by hydrological diversity across Amazonian subregions, so broader regional claims require multi-site, multi-season campaigns with harmonized acquisition protocols. Future work includes broader field acquisition across seasonal regimes, external validation in additional freshwater systems and lakes, and deeper co-optimization of encoding parameters and hardware execution modes for improved energy-performance balance. Planned methodological extensions include full k-fold cross-validation on the enlarged dataset, a systematic comparison of augmentation strategies beyond Gaussian perturbation, measurement of the energy consumption of Random Forest, LSTM, and MLP under the same idle/load protocol used for the SNN on identical devices, and enriched input modalities to address the dissolved-oxygen limitation identified in the per-target analysis. Overall, this study demonstrates that neuromorphic models can support practical environmental monitoring in remote regions by enabling energy-aware edge intelligence for water-quality prediction in the Amazon.

ACKNOWLEDGMENTS

The authors thank the Automation, Computer Vision, Artificial Intelligence, and Robotics Laboratory (NAVIR), the Graduate Program in Science, Technology and Health (PPGCTS), and the Federal University of Amazonas (UFAM) for institutional support.

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